Sharp equilibrium bounds for $(n, n, n-1)$ mass-action networks: an answer to a question of Banaji and Feliu

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Solver: Claude Opus 5.5 (effort xhigh), orchestrating agent · $n = 3$ case and computer searches: Claude Opus 5.5 exploration agents (effort high), 17 Oct 2026 · Independent verification: pending

Abstract. Banaji and Feliu proved that a mass-action network with n species, n reactions and rank $n-1$ has at most n positive nondegenerate equilibria on each stoichiometric class, and left open whether this bound is attained for general n and, if so, what source molecularity is needed. We construct, for every $n \geq 3$, an explicit $(n, n, n-1)$ network with sources X_1 and $2X_{i-1} + X_i$ that has exactly n positive nondegenerate equilibria on one stoichiometric class for suitable rate constants. Together with their bound of two equilibria for quadratic networks, this shows that the minimum source molecularity is exactly 3 for every $n \geq 2$. The proof is short and computer-free: the reaction vectors form a path through the species, the equilibria lie on a monomial curve, and the conservation law becomes a signomial with alternating coefficient signs.

The question

Banaji and Feliu [1] study positive equilibria of mass-action networks with n species, m reactions and stoichiometric rank r, called (n, m, r) networks. Their Corollary 4.21(i) states that $(n, n, n-1)$ networks admit no more than n positive nondegenerate equilibria on any stoichiometric class. In Remark 4.23 they write (arXiv:2409.06877v5, p. 30):

> *"In the case $n = 2$, both bounds in Corollary 4.21 are achieved by the $(2, 2, 1)$ network $X + 2Y \to 3Y$, $Y \to X$ [...]. We remark, however, that quadratic $(2, 2, 1)$ networks do not admit multiple positive nondegenerate equilibria [...]. We leave open whether the bound on positive nondegenerate equilibria in Corollary 4.21(i) is achieved for general n and, if so, the minimum source molecularity of a network needed to achieve the bound."*

We answer both parts.

Theorem 1. For every $n \geq 2$ there is an $(n, n, n-1)$ mass-action network with source molecularity 3 which, for suitable rate constants, has exactly n positive equilibria on some stoichiometric class, all of them nondegenerate. Hence the bound of Corollary 4.21(i) [1] is attained for every n.

Corollary 2. For every $n \geq 2$, the minimum source molecularity of an $(n, n, n-1)$ network attaining the bound n is exactly 3.

Proof (of Corollary 2 from Theorem 1)**.** For $n = 2$ this is the content of Remark 4.23 itself: the network $X + 2Y \to 3Y$, $Y \to X$ has source molecularity 3, and quadratic

$(2,2,1)$ networks admit at most one positive nondegenerate equilibrium per class. For $n \geq 3$, Corollary 5.16 of [1] states that quadratic $(n, n, n-1)$ networks (all sources of molecularity at most 2) admit no more than two positive nondegenerate equilibria on each stoichiometric class, fewer than n. Theorem 1 provides networks of source molecularity 3. □

1 Notation

Species are $X_1, \dots, X_n$ with concentrations $x \in \mathbb{R}^n_{>0}$. Reaction R_j has source complex $a_j \in \mathbb{Z}^n_{\geq 0}$, product complex $b_j \in \mathbb{Z}^n_{\geq 0}$ and reaction vector $\gamma_j = b_j - a_j$. We write $\Gamma = [\gamma_1 | \cdots | \gamma_m] \in \mathbb{Z}^{n \times m}$ for the stoichiometric matrix and $S \in \mathbb{Z}^{m \times n}$ for the matrix whose j-th row is a_j. Mass-action kinetics with rate constants $\kappa \in \mathbb{R}^m_{>0}$ gives

$$\dot{x} = \Gamma\, v(x), \qquad v_j(x) = \kappa_j\, x^{a_j} = \kappa_j \prod_i x_i^{a_{ji}},$$

with Jacobian $J(x) = \Gamma\, D_{v(x)}\, S\, D_{1/x}$, where D_u is the diagonal matrix with diagonal u. Following §2 [1], a positive equilibrium x is *nondegenerate* if $J(x)$ acts as a nonsingular transformation on $\operatorname{im}\Gamma$, i.e. $\ker J(x) \cap \operatorname{im}\Gamma = \{0\}$. Stoichiometric classes are the sets $(x_0 + \operatorname{im}\Gamma) \cap \mathbb{R}^n_{\geq 0}$. For $\lambda > 0$ and $q \in \mathbb{R}^n$ we write $\lambda^q = (\lambda^{q_1}, \dots, \lambda^{q_n})$; $\circ$ is the componentwise product and e_i the i-th unit vector.

2 The family of networks

Fix $n \geq 3$, define the signs

$$\varepsilon_i = (-1)^{\lfloor (i-1)/2 \rfloor} \quad (i = 1, \dots, n), \qquad \text{i.e. } \varepsilon = (+, +, -, -, +, +, -, -, \dots),$$

and $s = -\varepsilon_n \in \{\pm 1\}$. The network $\mathcal{N}_n$ consists of the n reactions

$$\begin{aligned}
R_1 &: \; X_1 \longrightarrow X_1 + \gamma_1, \\
R_{2k} &: \; 2X_{2k-1} + X_{2k} \longrightarrow 2X_{2k-1} + X_{2k} + \gamma_{2k} \qquad (2k \leq n), \\
R_{2k+1} &: \; 2X_{2k} + X_{2k+1} \longrightarrow 2X_{2k} + X_{2k+1} + \gamma_{2k+1} \qquad (2k+1 \leq n),
\end{aligned}$$

with reaction vectors

$$\begin{aligned}
\gamma_1 &= s\, e_1 + e_n, \\
\gamma_{2k} &= c_k\, (e_{2k} - e_{2k-1}), \quad c_k = (-1)^{k+1} s, \\
\gamma_{2k+1} &= d_k\, (e_{2k} + e_{2k+1}), \quad d_k = (-1)^k s.
\end{aligned}$$

Written out, the reactions are listed in Table 3.

reaction	coefficient $+1$	coefficient -1
R_1 (s)	$X_1 \to 2X_1 + X_n$	$X_1 \to X_n$
R_{2k} (c_k)	$2X_{2k-1} + X_{2k} \to X_{2k-1} + 2X_{2k}$	$2X_{2k-1} + X_{2k} \to 3X_{2k-1}$
R_{2k+1} (d_k)	$2X_{2k} + X_{2k+1} \to 3X_{2k} + 2X_{2k+1}$	$2X_{2k} + X_{2k+1} \to X_{2k}$

Table 3 / The reactions of $\mathcal{N}_n$.

All products are nonnegative; the sources are $a_1 = e_1$ and $a_j = 2e_{j-1} + e_j$ $(2 \leq j \leq n)$, so the source molecularity is 3 and the product molecularity is at most 5. For instance

$$\mathcal{N}_3 : \; X_1 \to 2X_1 + X_3, \quad 2X_1 + X_2 \to X_1 + 2X_2, \quad 2X_2 + X_3 \to X_2;$$
$$\mathcal{N}_4 : \; X_1 \to X_4, \quad 2X_1 + X_2 \to 3X_1, \quad 2X_2 + X_3 \to 3X_2 + 2X_3, \quad 2X_3 + X_4 \to X_3 + 2X_4.$$

Lemma 4 (stoichiometry). $\mathcal{N}_n$ is an $(n, n, n-1)$ network with $\operatorname{im}\Gamma = \varepsilon^{\perp}$ (so $\ker\Gamma^{\mathsf{T}} = \operatorname{span}(\varepsilon)$) and $\ker\Gamma = \operatorname{span}(\mathbf{1})$, $\mathbf{1} = (1, \dots, 1)$. In particular it is dynamically nontrivial and its stoichiometric classes are the sets $\{x \geq 0 : \varepsilon \cdot x = T\}$, $T \in \mathbb{R}$.

Proof. For $t = 1, \dots, n-1$ let $u_t = e_{t+1} - e_t$ if t is odd and $u_t = e_t + e_{t+1}$ if t is even. Then γ_{t+1} is a nonzero multiple of u_t ($\gamma_{2k} = c_k u_{2k-1}$, $\gamma_{2k+1} = d_k u_{2k}$). If t is odd, $\varepsilon_t = \varepsilon_{t+1}$; if t is even, $\varepsilon_{t+1} = -\varepsilon_t$. Hence $\varepsilon \cdot u_t = 0$ for all t; also $\varepsilon \cdot \gamma_1 = s + \varepsilon_n = 0$. The vectors $u_1, \dots, u_{n-1}$ are linearly independent: deleting the first coordinate leaves an upper bidiagonal $(n-1) \times (n-1)$ matrix with unit diagonal (u_t is supported on $\{t, t+1\}$ with coefficient 1 at $t+1$). Therefore $\operatorname{im}\Gamma = \operatorname{span}(u_1, \dots, u_{n-1}) = \varepsilon^{\perp}$ and $\operatorname{rank}\Gamma = n - 1$.

We check $\sum_j \gamma_j = 0$ coordinatewise. Coordinate 1: $s - c_1 = 0$. Coordinate $2k$: $c_k + d_k = 0$ if $2k < n$; if $2k = n$ the coordinate is $c_k + 1$, and $s = -\varepsilon_{2k} = -(-1)^{k-1}$ gives $c_k = -1$. Coordinate $2k+1 \geq 3$: $d_k - c_{k+1} = (-1)^k s - (-1)^{k+2} s = 0$ if $2k+1 < n$; if $2k+1 = n$ the coordinate is $d_k + 1$, and $s = -\varepsilon_{2k+1} = -(-1)^k$ gives $d_k = -1$. So $\Gamma\mathbf{1} = 0$, and since $\operatorname{rank}\Gamma = n - 1 = m - 1$, $\ker\Gamma = \operatorname{span}(\mathbf{1})$. □

Lemma 5 (exponents). The source matrix S is lower bidiagonal with unit diagonal, hence invertible, and $q := S^{-1}\mathbf{1}$ has entries

$$q_i = \frac{1 + 2(-2)^{i-1}}{3} \qquad (i = 1, \dots, n), \qquad q = (1, -1, 3, -5, 11, -21, 43, \dots).$$

These are pairwise distinct nonzero integers, ordered as $q_{2\lfloor n/2 \rfloor} < \cdots < q_4 < q_2 < 0 < q_1 < q_3 < q_5 < \cdots$.

Proof. $Sq = \mathbf{1}$ reads $q_1 = 1$ and $2q_{i-1} + q_i = 1$, i.e. $q_i - \frac{1}{3} = -2(q_{i-1} - \frac{1}{3})$. For odd i, $q_i = (1 + 2^i)/3 > 0$ increases with i; for even i, $q_i = (1 - 2^i)/3 < 0$ decreases with i. □

3 Equilibria lie on a monomial curve

Lemma 6. Fix $\kappa \in \mathbb{R}^n_{>0}$ and put $C = \exp(-S^{-1} \log \kappa) \in \mathbb{R}^n_{>0}$ (componentwise). The positive equilibria of $\mathcal{N}_n$ are exactly the points $x(\lambda) = C \circ \lambda^q$, $\lambda > 0$, and at $x(\lambda)$ every reaction has rate $v_j = \lambda$. Conversely, every $C \in \mathbb{R}^n_{>0}$ arises from the rate constants $\kappa_j = C^{-a_j}$.

Proof. By Lemma 4, $x > 0$ is an equilibrium iff $v(x) \in \ker\Gamma = \operatorname{span}(\mathbf{1})$, i.e. $\kappa_j x^{a_j} = \lambda$ for all j and some $\lambda > 0$. Taking logarithms, $S \log x = (\log\lambda)\mathbf{1} - \log\kappa$, so $\log x = (\log\lambda)\, q + \log C$. The converse follows from $\log\kappa = -S \log C$. □

On the class $\varepsilon \cdot x = T$, positive equilibria therefore correspond bijectively ($\lambda \mapsto x(\lambda)$ is injective because $q \neq 0$) to the positive zeros of

$$F(\lambda) = \sum_{i=1}^{n} \varepsilon_i C_i \, \lambda^{q_i} - T.$$

Lemma 7 (nondegeneracy). The equilibrium $x(\lambda)$ is nondegenerate if and only if $F'(\lambda) \neq 0$.

Proof. At $x = x(\lambda)$ we have $v = \lambda \mathbf{1}$, so $J = \lambda \Gamma S D_{1/x}$. Then $Jy = 0$ iff $SD_{1/x}y \in \ker \Gamma = \operatorname{span}(\mathbf{1})$ iff $D_{1/x}y \in \operatorname{span}(S^{-1}\mathbf{1}) = \operatorname{span}(q)$ iff $y \in \operatorname{span}(x \circ q)$. Since $x'(\lambda) = \lambda^{-1}\, x(\lambda) \circ q$, we get $\ker J = \operatorname{span}(x'(\lambda))$. Because $J(\mathbb{R}^n) \subseteq \operatorname{im} \Gamma = \varepsilon^{\perp}$, $\ker J \cap \operatorname{im} \Gamma = \{0\}$ iff $x'(\lambda) \notin \varepsilon^{\perp}$ iff $\varepsilon \cdot x'(\lambda) = F'(\lambda) \neq 0$. □

4 Signomials with prescribed roots

Lemma 8 (Descartes' rule for real exponents)**.** Let $e_0 < e_1 < \cdots < e_N$ be real and $c \in \mathbb{R}^{N+1} \backslash \{0\}$. The number of zeros of $f(t) = \sum_k c_k t^{e_k}$ in $(0, \infty)$, counted with multiplicity, is at most the number of sign changes of $(c_0, \ldots, c_N)$ (zero entries ignored).

This is classical (Laguerre; see Part V, Ch. 1 [2] or [3]).

Proof (sketch)**.** Induct on the number V of sign changes. If $V = 0$, f has no positive zero. Otherwise pick θ strictly between two consecutive exponents across which the sign changes, and consider $g(t) = (t^{-\theta} f(t))' = \sum_k c_k (e_k - \theta) t^{e_k - \theta - 1}$. Multiplication by $e_k - \theta$ flips exactly the signs on one side of the chosen change, so g has $V - 1$ sign changes, while by Rolle's theorem g has at least $Z(f) - 1$ positive zeros counted with multiplicity. □

Lemma 9. Let $e_0 < \cdots < e_N$ be real and $0 < \lambda_1 < \cdots < \lambda_N$. There is a nonzero $c \in \mathbb{R}^{N+1}$ with $f(\lambda_r) = \sum_k c_k \lambda_r^{e_k} = 0$ for all r. Every such c has all entries nonzero and strictly alternating in sign, and $\lambda_1, \ldots, \lambda_N$ are then all the positive zeros of f, each of them simple.

Proof. Existence: N homogeneous linear equations in $N + 1$ unknowns. By Lemma 8, f having at least N positive zeros forces at least N sign changes in $(c_0, \ldots, c_N)$; with $N + 1$ entries all of them are nonzero and they alternate. Lemma 8 again shows that f has at most N positive zeros counted with multiplicity, so the λ_r are all of them and each is simple. □

5 Proof of the main theorem

Proof (of Theorem 1)**.** The case $n = 2$ is in Remark 4.23 [1]. Let $n \geq 3$, choose $0 < \lambda_1 < \cdots < \lambda_n$, and list the $n + 1$ distinct exponents $\{q_1, \ldots, q_n, 0\}$ in increasing order (Lemma 5): $q_{2\lfloor n/2 \rfloor} < \cdots < q_4 < q_2 < 0 < q_1 < q_3 < \ldots$. Lemma 9 gives a strictly alternating coefficient vector c whose signomial vanishes exactly at the λ_r, all simple; replacing c by $-c$ if necessary, the coefficient of λ^{q_1} is positive. Read the signs ε_i in the same order: $\varepsilon_{2k} = (-1)^{k-1}$ alternates along the negative exponents $q_2, q_4, \ldots$, $\varepsilon_{2k+1} = (-1)^k$ alternates along the positive exponents $q_1, q_3, \ldots$, and $\varepsilon_1 = \varepsilon_2 = +1$. So the sequence of the ε's, with an entry -1 inserted at exponent 0, alternates and starts at λ^{q_1} with $+$, exactly like c. Hence the coefficient $c_{(q_i)}$ of λ^{q_i} has sign ε_i and the constant coefficient $c_{(0)}$ is negative. Define

$$C_i = \varepsilon_i\, c_{(q_i)} > 0, \qquad T = -c_{(0)} > 0, \qquad \kappa_j = C^{-a_j}.$$

By Lemma 6, the positive equilibria of $\mathcal{N}_n$ with these rate constants on the class $\varepsilon \cdot x = T$ correspond to the positive zeros of $F = f$, namely $\lambda_1, \ldots, \lambda_n$. They are simple, so the n equilibria $x(\lambda_r)$ are nondegenerate by Lemma 7, and there are no others on that class. □

Remark 10. **(1)** If the λ_r are rational, c can be chosen rational, so C, T, κ and the equilibria $x(\lambda_r) = C \circ \lambda_r^q$ are rational (the exponents are integers).

(2) The stoichiometric classes of $\mathcal{N}_n$ are unbounded (ε has entries of both signs), as they must be by Corollary 4.21(ii) [1] once $n \geq 3$.

(3) $\mathcal{N}_n$ is an $(n, n, n-1)$ network with trivial partition, so it also shows that the bound of Theorem 4.20(i) [1] is attained for every n when $m = n$.

(4) The exponents q_i grow like 2^i, which makes the rational witnesses large. Computer searches found other networks with linearly growing exponents, but without a uniform proof.

⌟

6 A small explicit instance

For $n = 3$ take $\lambda_r = r$. The exponents are $q = (1, -1, 3)$ and $\lambda F(\lambda) = -C_3\lambda^4 + C_1\lambda^2 - T\lambda + C_2$ must equal $-C_3(\lambda-1)(\lambda-2)(\lambda-3)(\lambda+6) = -C_3(\lambda^4 - 25\lambda^2 + 60\lambda - 36)$. With $C_3 = 1$: $C = (25, 36, 1)$ and $T = 60$. The network

$$X_1 \xrightarrow{1/25} 2X_1 + X_3, \qquad 2X_1 + X_2 \xrightarrow{1/22500} X_1 + 2X_2, \qquad 2X_2 + X_3 \xrightarrow{1/1296} X_2$$

has, on the class $x_1 + x_2 - x_3 = 60$, exactly the three positive equilibria $(25, 36, 1)$, $(50, 18, 8)$ and $(75, 12, 27)$. The Jacobian restricted to $\operatorname{im}\Gamma$ has determinant $7/450$, $-1/225$ and $1/150$ respectively, with eigenvalues $\{-1.053, -0.0148\}$, $\{-0.412, +0.0108\}$ and $\{-0.384, -0.0174\}$: two stable equilibria and a saddle.

7 Computer verification

The proof above needs no computer. As an independent consistency check, `scripts/certify_family.py` builds $\mathcal{N}_n$ for $n = 3, \ldots, 12$ and verifies in exact rational arithmetic (python-flint): the rank and kernels of Γ; the exponent formula; the strict sign alternation of the generalized-Vandermonde kernel vector for $\lambda_r = r$; the positivity of C, T and κ; that the n points $x(r)$ satisfy $\Gamma v(x) = 0$ and $\varepsilon \cdot x = T$ exactly; that the restricted Jacobian has nonzero determinant at each of them; and that F has exactly n sign changes (301 checks, all pass). `scripts/example_n3.py` checks the instance of the previous section with sympy.

References

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